Real Time Video Stabilization Methods in IR Domain for UAVs – A Review

Adeel Yousaf
Department of Avionics Engineering
Institute of Space Technology
Islamabad, Pakistan
adeelyousaf1993@gmail.com

Khurram Khurshid
Department of Electrical Engineering
Institute of Space Technology
Islamabad, Pakistan
khurram.khurshid@ist.edu.pk

Muhammad Jaleed Khan
Department of Electrical Engineering
Institute of Space Technology
Islamabad, Pakistan
mjk093@gmail.com

Muhammad Shehzad Hanif
Department of Electrical and Computer Engineering
King Abdul-Aziz University
Jeddah, KSA
mshehzadhanif@gmail.com

***Abstract*— In applications that involve a camera attached to a moving platform, such as UAV, Car, Drones etc., mechanical vibrations of the moving platform cause undesired shaking motion in videos known as Video Jittering. This causes severe performance degradation in applications such as object detection and tracking for applications such as surveillance and remote sensing. Video stabilization is required as preprocessing step to remove undesired motion while preserving the desired ones. Video Stabilization is an active research area for past decade and most of work done is on offline methods for stabilizing video in visible range. However stabilizing Infrared videos and that too for real time applications is still an open research area. This is because of the difficulty in detecting and tracking good features in IR videos due to lower resolution/quality videos. In this paper various existing methods provided in literature are compared in detail to focus on method that is more feasible for real time IR stabilization implementation.**

Keywords—Video Stabilization, Image Stabilization, Infrared Videos, Real Time Stabilization, Motion Vector, Feature points, Image Wrapping, Camera Paths, Image compensation, Motion estimation

I. INTRODUCTION

In this digital world, camera based applications had grown its deep roots. Most of these applications are of sensitive nature for example Video surveillance [36, 37, 38], object tracking [39, 40, 41] .etc. that need accurate and stable frames at input for further processing. In context of video stabilization we can divide such applications in two categories i.e. camera mounted on Non-moving platforms and camera mounted on moving platforms. Latter one includes camera connected with Unmanned Air Vehicles (UAV's), Cars, Drones etc. due to their mechanical vibrations undesired shaking motion is induced in videos, commonly known as Video Jittering. Video Jittering not only results in quality degradation plus it also decreases the efficiency of further applied applications like video surveillance, object tracking, video compression etc. To remove these high frequency motions from video sequence, video stabilization is applied. Number of video stabilization techniques [42, 58] have been introduced that can be broadly classified into three categories a) Digital Image stabilization b) Mechanical Image stabilization c) Optical image stabilization**.** In early days by using sensors like accelerometers and vibrational feedback mechanical image stabilization is achieved, then it came era of optical stabilization. In this scheme movable lens is used, that changes the path of light as light passes thorough camera. Both of these techniques have enormous cost related with them as both of them are hardware dependent. These schemes are not suitable for small camera modules such as in cellphones. Digital image stabilization (DIS) applies digital image processing algorithms to smooth the shaking video sequence. DIS can also be implemented along with Mechanical and Optical techniques to enhance the results. Depending upon processing time DIS can be classified in online and offline categories. In offline technique one has access to all future frames i.e. whole video sequence is stored first and then video stabilization algorithms are applied, in online real time stabilized image sequence is required at output as frames with undesired motions are received at input. Video stabilization consists of three fundamental steps [43] Motion Estimation, Motion Correction (Smoothing) and Frame Compensation as shown in ''*Fig* 1''.

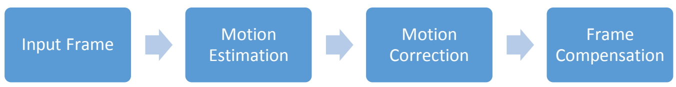

Fig 1. Digital image stabilization system basic scheme

Motion estimation calculates the global motion between two consecutive frames. Often Global Motion Vectors contains some outlier's i.e. inaccurate motion vectors, several techniques in literature are proposed to avoid these misleading motion vectors, and it includes RANSAC [44], LMS, and MSAC etc. After motion vector estimation motion correction differentiate the intentional and unintentional motions using methods like Motion vector Integration [45]; Low pass filtering, Kalman Filters etc. At the end is Frame compensation, frame is wrapped by inverse of unintentional motion, so it appears smooth. Out of these three steps motion estimation is the foremost important step as its output is the input to further two blocks, so overall efficiency of stabilization is highly dependent upon Motion Estimation. In this step motion parameters are estimated and then forwarded to motion correction. Depending upon Motion Models (MM) video stabilization can be classified as 2D, 2.5D and 3D methods. Starting from 3D methods [20, 24, 25, 26, 27, 28, 29, 30] they perform well against video containing different depth scenes (Parallax), however 3D method are least robust, as these method don't perform well on frames with no depth variation, addionaly they require Structure from motion [23] which involves large number of feature points and is very computationally expensive i.e. it's not suited at all for real time implementation. 2D method [21, 22, 31, 32, 33, 34, 35] uses Homography (Affine motion model etc.) for stabilizing videos. Homography is valid only for planer scenes, when scene has high depth variation 2D method becomes invalid. But 2D methods are most robust and computationally least expensive methods, most suited for real time implementation. Different techniques are presented in literature to make 2D method work as well as 3D method along with robustness of 2D methods. 2.5D method [17, 18] actually fills the gaps between 3D and 2D methods by relaxing 3D reconstruction requirement. These methods require long trajectories, acquiring such long trajectories is not possible in videos containing fast pane i.e. fast horizontal motion as objects enter and leave the scene so getting long trajectory is not possible. To achieve real time performance as desired, our survey mainly focuses on 2D techniques. Under 2D transformation lie affine and projective transformation models. Affine transform preserves parallelism and mid points, it helps in catering rotation, scaling and shear between images. On other hand projective transform is more general then affine; sometimes due to perspective projection parallelism is not preserve i.e. parallel lines intersect at the end. In such case affine model is of no use so here projective motion model is used.

$$p' = Hp \tag{1}$$

$\boldsymbol{H}$ is 3x3 Homography matrix also known as fundamental matrix, where $\boldsymbol{p}$ and $\boldsymbol{p'}$ are pixel coordinates in adjacent frames. Motion estimation step basically find H matrix parameters. Apart from 2D and 3D motion model there exists other models known as Cylindrical and Spherical Coordinates, It's of very limited practical use as information about camera tilt angle is required for calculation of motion parameters, which is mostly not available. After selection of suitable motion model, motion estimation calculates its unknown parameters. Motion estimation can be achieved using direct pixel based or feature based approaches. In direct methods pixel to pixel matching is used. For matching error metric is used. In literature various such metrics are proposed like SSD, SRD, SAD, MAD or correlation. Each has its own pros and cons. As direct methods use pixel to pixel matching that's why they have high computational load. To optimize this problem Hierarchal motion model is used. Basically image pyramid is constructed and matching is done from coarser to fine level. In certain scenarios it is not possible to coarsen the image as information may decrease below certain level and image may get blur. So Fourier Transform is used as an alternative. Basically Fourier Transform of shifted signals has linearly varying phase and same magnitude as original. Windowed Correlation is used in some cases in place of Fourier convolution when overlap between adjacent frames is very small. Some authors have proposed phase correlation for estimating unknown parameters. Phase correlation results in impulse (Peak), relating values of shift between images. Performance of phase correlation method highly depends upon signal to noise ratio. Gradient cross correlation has been proposed as an alternative to phase correlation. Initially Fourier based techniques are only used for finding translation but later techniques have been proposed using polar coordinates through which rotation and scaling can also be estimated. Apart from direct pixel based matching, feature based motion estimation is widely used. Basically key points are tracked in adjacent frames, and depending upon the motion of these descriptors motion model parameters is estimated. Classical Video Stabilization (Feature Based) achieves reasonable stabilized output for frames with Low Noise, High Contrast Level and Less Blur i.e. images in which reasonable number of features can be tracked thorough out the video sequence. Videos Captured in visible ranges satisfies these conditions but IR sensors have low sensitivity as compared with CMOS and CCD used in visual images. Due to less sensitivity images captured have low contrast level and are mostly blurring. Plus they have lower resolution as compared with visual images. Due to these factors finding reasonable numbers of features in IR videos is quite challenging. In this paper detailed review of existing video stabilization techniques is presented. In section 2, various kinds of motion models are presented, followed by detail explanation and comparison of existing methods. Finally it closes with open discussion of limitations of existing stabilization techniques.

II. MOTION MODELS

Pixels in consecutive frames are related mathematically by motion models. Number of such models ranging from simple 2D to complex 3D are presented in literature [49, 50, 51]. As

3D motion models are very complex and are not suitable for strict real time scenarios, this paper mainly focuses on 2D motion models. 2D motion models are for planar scenes i.e. scene with small or no depth change. 2D motion model includes Translation, Rotation plus Translation, Scaled Rotation, Affine and Projective motion models.
2D Translation model can be expressed as

$$x' = [I \quad t]\, x \tag{2}$$

Where $\boldsymbol{x}$ is pixel coordinates in frame $(\boldsymbol{t})$, $\boldsymbol{x}'$ are pixel coordinates in frame $(\boldsymbol{t+1})$. $\boldsymbol{I}$ is identity matrix.

2D Translation plus rotation is also known as rigid motion model and it can be expressed as,

$$x' = [\,R \quad t\,]\, x \tag{3}$$

Where $\boldsymbol{R}$ is rotation matrix.
2D Affine motion model can be expressed as

$$\mathrm{x}' = \begin{bmatrix} \mathrm{a}_{00} & \mathrm{a}_{01} & \mathrm{a}_{02} \\ \mathrm{a}_{10} & \mathrm{a}_{11} & \mathrm{a}_{12} \end{bmatrix} \mathrm{x} \tag{4}$$

2D Projective motion model is also known as Homography or perspective projection it can be expressed as,

$$x' = \begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{bmatrix} x \tag{5}$$

In Affine transformation origin does not necessarily map to origin but it preserves parallel lines and ratios, it contains 6 parameters. On other hand projective transformation maps lines to lines but does not necessarily preserves parallelism, it comprises of 8 parameters. In motion models there exists a hierarchy; each simpler model is subset of a more complex model. Hierarchy of 2D motion model is provided in table 1.

TABLE 1. HIERARCHY OF 2D COORDINATES TRANSFORMATIONS

Name	Matrix	DOF
Translation	$[\mathrm{I} \mid \mathrm{t}\,]_{2x3}$	2
Rigid	$[\mathrm{R} \mid \mathrm{t}\,]_{2x3}$	3
Similarity	$[\mathrm{sR} \mid \mathrm{t}\,]_{2x3}$	4
Affine	$[A\,]_{2x3}$	6
Projective	$[H\,]_{3x3}$	8

III. LITERATURE REVIEW

To estimate inter frame motion Battiato et al. [1] extracted Scale Invariant Feature Transform (SIFT) features and tracked them in consecutive frames. Advanced Least Square Method [47] is implemented to avoid estimation errors. Motion Vector Integration (MVI) is employed to separate intentional motion from unintentional motion as Kalman filter increases the computational load, which leads to opposition in real time implementation. SIFT features [48] are known for extracting highly distinctive information known as descriptors from images. SIFT key points are extracted from two consecutive frames to find the local motion vector. It is to be noted here that not all of the local motion vectors provide correct information about how these two frames have moved with respect to each other for example due to presence of large moving object. To discard these false key points euclidian distance between descriptors is employed. Threshold is set there to discard misleading key points. Authors found that selecting 0.7 as threshold performs better in discarding the wrong matches. In this paper Linear Conformal Model (Eqn. 6) is used that handles Translation (T_X, Ty), Rotation and zoom factor. To find these four parameters only four equations i.e. two couples of features are required, but it is quite possible that features are affected with noise so Least Square Method is employed to find these parameters.

$$\begin{aligned} x_f &= x_i\,\gamma cos\theta - y_i\,\gamma sin\theta + T_x \\ y_f &= x_i\,\gamma sin\theta + y_i\,\gamma cos\theta + T_y \end{aligned} \tag{6}$$

Where $\boldsymbol{\gamma}$ is zoom parameter.

As jerks in video may not always be there due to camera shake but it can be due to Intentional motion. So to separate these both MVI is applied. Its equation is provided below.

$$IMV(n) = \alpha IMV(n-1) + GMV(n) \tag{7}$$

Where $\boldsymbol{\alpha}$ is damping factor its value ranges from 0-1. Choosing high value will result in low jitter but could also effect in following intentional motion and vice versa. Instead of selecting a fixed value adaptive damping factor is used by authors. Peak Signal to Noise Ratio (PSNR) is selected as a measure for evaluating stabilization performance. PSNR between consecutive frames can be defined as

$$PNSR\ (t) = 10log_{10}\, \frac{I^2 max}{MSE\ (t)} \tag{8}$$

Where MSE is Mean Square Error and I_{max} is the maximum intensity. PNSR is measure of similarity between consecutive frames, as after stabilization frames become more continuous so PNSR increases as shown in figure 2. SIFT features are employed by authors, which have high computational complexity due to which this method is not suitable for real time implementation.

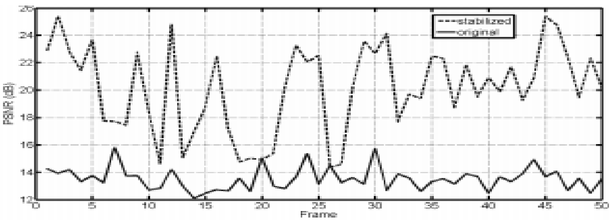

Fig 2. PNSR values for Original and Stabilized sequence [1]

Corsini and Giovanni in [3] have compared three different methodologies in terms of stabilized output and computational load to determine the best method for real time implementation. Planar motion model is selected by authors for further evaluation and for comparison three methodologies i.e. Differential Techniques, Phase Correlation and Hierarchical Block Matching are selected. Derivative of image is employed in diffenrentional techniques to estimate the optical flow. Fourier Transform is applied in Phase Correlation technique and then Translation is estimated by evaluating the peak in Cross power Spectrum. In Block Matching techniques, frame is subdivided in blocks and then corresponding blocks are matched in consecutive frames within the search window. It ignores the rotational motion and handles translational motion. Three level pyramidal decomposition of frame are done to reduce the computational load. In case of large moving objects in scene, performance of phase correlation degrades while other techniques are not affected by these phenomena. On basis of computational load hierarchical block matching techniques performs well. Performance of techniques highly depends upon the scenario in which they are being applied. Comparison of computational loads of above mentioned techniques is provided in table 2.

TABLE 2. COMPUTATIONAL LOAD FOR ANALYZED MOTION ESTIMATION TECHNIQUES

Technique	**Number of Operations**
Hierarchical Block Matching	9.28 * R * C
Differential Techniques	$O(R * C)^4$
Phase Correlation	R * C * (4*log(R*C) +1)

In [4] authors have presented a technique to increase the performance of 2D motion model against videos containing parallax. KLT feature tracker [46] is used, feature belonging to moving objects in scenes or features that get occluded are discarded at initial stage and remaining feature trajectories are filtered to generate smooth trajectory. $\boldsymbol{N}$ number of features are tracked in consecutive frames. $\boldsymbol{N}$ is selected depending upon the resolution of video. Feature trajectories that are not lost for $\boldsymbol{\alpha}$ frame are selected and rest is discarded in order to avoid trajectories of moving objects that only last for small number of frames. Initially $\boldsymbol{\alpha}$ is selected as total number of frames and then Gradient Descent Method (GDM) is applied until all of features trajectories spanning $\boldsymbol{\alpha}$ frames are at least 0.5N [Eqn. 9].

$$\alpha = \alpha - \epsilon(0.5\ \mathrm{N} - \mathrm{k}) \qquad (9)$$

$\boldsymbol{k}$ is number of trajectories being tracked in $\boldsymbol{\alpha}$ frames and $\boldsymbol{\epsilon}$ is the update rate. This adaptive selection of $\boldsymbol{\alpha}$ allows video stabilization to recover even after not having long trajectories. After getting enough long trajectories low pass filter is applied to remove high frequency shaking motion. In this paper authors have used affine transformation $\boldsymbol{H_j}$ as motion model.

$$\mathrm{H_j} = \begin{bmatrix} \mathrm{scos}(\theta) & -\mathrm{ssin}(\theta) & \mathrm{Tx} \\ \mathrm{ssin}(\theta) & \mathrm{scos}(\theta) & \mathrm{Ty} \\ 0 & 0 & 1 \end{bmatrix} \qquad (10)$$

Random Sample Consensus (RANSAC) is used to find the unknown parameters in above provided equation [Eqn. 10]. RANSAC works well in discarding the outliers. 2D motion model actually depends upon pattern of major content in video sequence and hence in case of parallax produce very weak results. In [4] clustering is used to handle this type of motion. Basically trajectories that experience similar jitter are said to belong to approximately equal depth, they are grouped to one cluster. Each cluster is smoothened independently. Those Feature trajectories which lie into no cluster are discarded and they are known as erroneous trajectories. At the end non erroneous transformed trajectories are used for wrapping the frame. Detailed experimental analysis has been provided by authors. Proposed technique have been compared with iMovie [5], Deshakar [6] and with latest work at that time in panoramic video stabilization [7]. Proposed technique has been tested on around 40 videos and works well on video sequence with long trajectories, in one of video sequence named as “Speed breaker”, it’s captured from a moving vehicle, there are some frames in which electric pole passes in front of camera with relative less distance as compared with other objects. Proposed technique was not able to stabilize these frames but after pole passes, again quite good stabilization was achieved due to robustness of equation 9.

In [8, 9] authors provides real time video stabilization for fast vehicles to cater high vibrating images. Their technique comprises of four steps that includes Frame Shaking Judgment, Feature Classification, Global Motion and Rotation Angle Calculation and Frame Compensation, its flow chart is provided in figure 3. Harris corner points are detected and tracked by using LK optical flow method in consecutive frames. In the very first step i.e. frame shaking judgment, feature points and their motion vectors are used for deciding either frame is shaking or not.

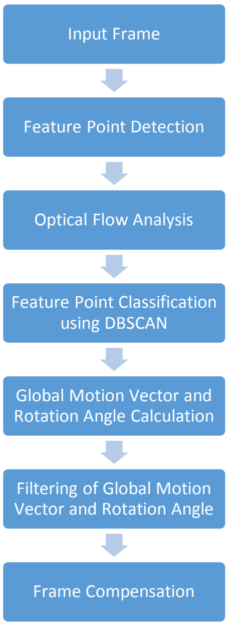

Fig 3. Flow chart of proposed algorithm [8, 9]

When frame is shaking, the most of the optical flow vectors are concentrated in that direction. Rotation angle histogram of optical flow vectors are calculated and if probability of maximum rotation angle is greater than specific threshold then frame is considered as shaking. In second step i.e. feature classification, features are classified as background or foreground depending upon the distance between feature points in consecutive frames. Basically four feature points are selected at a time and homography matrix is calculated. Then this specific homography matrix is applied to Frame (T-1) and overlapping features are calculated between frame (T) and transformed frame (T-1), this process is repeated by selected next four feature points. At the end homography matrix that results in highest number of overlapping features points is selected and applied to Frame (T-1). Features points with zero distance are marked as background other are categorized as foreground.

To calculate global rotation and motion Hough transform and DBSCAN clustering algorithem is used, and cluster with maximum number of motion vector is selected.

Detailed comparison is provided with Deshakar (Publically available tool) and with 3D video stabilization technique i.e. spatially and temporally optimized method. 3D technique offers best results but its 22 -27 times more computationally expensive then proposed technique and it's not suitable for real time applications. Deshakar is 2D video stabilization method produce very weak results on test bench video sequence. When it comes to real time implementation and acceptable video stabilization results this method is up to work.

In [10] authors have replaced feature trajectories with pixel profiles. Video sequence for which traditional feature matching algorithms and 3D reconstruction is not possible due to absence of long feature trajectories, can be stabilized using pixel profiles. Authors have used steady flow which is modified version of optical flow to identify discontinuous motion using spatial temporal analysis, although motion segmentation techniques [52, 53, 54, 55] can be used for this purpose but they usually require long feature trajectories. This method achieves noble performance but for offline applications as its iterative process.

Various stabilization methods that uses phase correlation for motion estimation are provided in literature to stabilize image sequence [56, 57]. In [11] phase correlation of central part of frame is used and it is assumed that camera is placed on front of a moving vehicle. As vehicle moves forward background moves in opposite direction, different regions of image moves at different velocities i.e. object near camera moves faster as compared to objects at long distance from camera. Optical flow of such scenario resembles that of

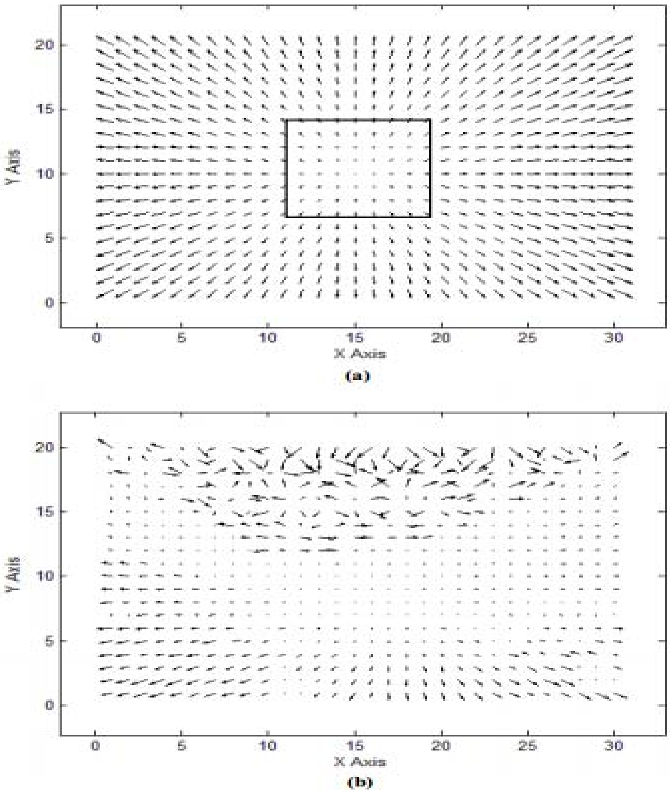

Fig 4. (a) Optical flow of ideal camera zooming (b) Actual optical flow of vehicle moving forward [11]

camera optical zooming as shown in respective figure 4(a). Central region having optical flow with small length represents the region far from camera and vice versa. Figure 4(b) represents actual optical flow. According to authors global motion estimation will be more accurate if such area out of frame is selected that has small intentional zooming, due to this assumption central part of frame is selected. Motion vector that maximizes the phase correlation between consecutive frames is selected as Global motion vector. Rotation is not dealt by authors in this paper.

In [12] authors have achieved video stabilization using classification of feature points as foreground and background. Feature matching based techniques perform better in case of scale, illumination and view point changes, but miscorrespondence and presence of moving objects degrades its performance, authors have presented a novel technique for motion estimation using only background feature points. perspective projection is used to describe global motion between two consecutive frames. Initially four feature points are selected and then using RANSAC fundamental matrix is calculated, feature points are transformed using fundamental matrix and distance is calculated using L2 Norm between initial feature points and transformed feature points. Set of feature points correspondences which satisfies the threshold are determined. Above steps are repeated for ***K*** times. And fundamental matrix which provides maximum number of feature correspondence is selected. For estimation of intentional motion Kaman filter is used.

In [13] authors have presented block based fast video stabilization technique. Circular blocks are used here as shown in figure 5 because they are more robust then rectangular blocks against rotation. After obtaining motion vectors a polynomial fitting and predicting method (PFPM) is employed to differentiate desired and undesired motion vectors. Image compensation cannot be applied directly to motion vectors obtained at very first step, desired and undesired motion vectors have to be differentiated. Usually desired motion is smooth and slow while undesired is fast and random. Low pas filtering is applied to keep low pass motion and reject high pass motion but this method introduces undesired results as stabilized video lags behind original by several frames.

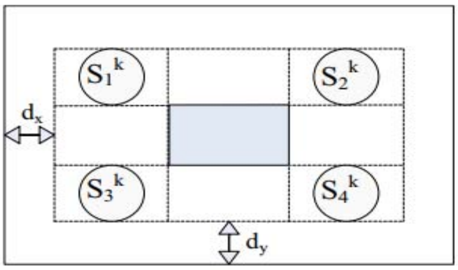

Fig 5. Selection of matching blocks in current frame [13]

Particle filters are commonly used in video tracking [59, 60, 61] and pose estimation. In [14, 16] authors have employed it in video stabilization. Initially SIFT based feature matching technique is used. SIFT method might get wrong feature points, but still particle filter performs well as particle filter relies on samples around the SIFT output rather than output itself. Constant velocity Kalman Filter model is used to estimate intentional camera movements.

Short comings of feature based techniques are presented in [15]. As per authors optical flow techniques use first order approximation of Taylor series, which mean that motion cannot be very large. Otherwise higher terms of Taylor series will be ignored and it will lead to large errors. To overcome this pyramid based techniques are used due to which large translation motion are scaled to small translation motion in lower resolution images, but it cannot tackle large rotational motion because the rotational motions in highest and lowest resolution is same. To overcome this authors have used circular blocks in consecutive frames to find the fundamental matrix.

In [17] authors have achieved video stabilization by implementing subspace constraints on feature trajectories while smoothing them. Authors have highlighted the limitations of 2D video stabilization technique plus they have also mentioned problems incurred by 3D video stabilization methods. This paper focuses on second step of stabilization i.e. smoothing feature trajectories. 4 step method is presented by authors to achieve smoothness's. This method performs well on videos in which long trajectories can be tracked. But tracking long feature trajectories is not possible in videos involving panning such as captured from moving vehicles. To achieve high performance along with avoiding computational cost of 3D methods authors in [18] have used projective reconstruction to stabilize jittered video. It's an intermediate method between 2D and 3D techniques. Providing literature review of previous techniques author states that 2D techniques are very efficient and very robust when scenes are planar and jitter is due to pure rotation, but when these conditions are not met then 2D methods are not sufficient. On other hand 3D methods are very efficient in handling parallax but they are very computationally expensive, moreover they require large number of feature points for tracking. Model presented by authors don't recover 3D camera pose and 3D locations, and only calculate geometric relations between feature points and Epipolar lines. In this way this method avoids the computational burden of 3D techniques. Initially virtual point trajectories are calculated using KLT [19] tracker and then these trajectories are smoothed using Gaussian kernel. Then correspondence between original jittered trajectory and smoothed trajectory is calculated to find fundamental matrix. Finally these matrices are used to wrap the frames. Authors have also highlighted the limitation of this method, scenes with excessive jitter cannot be tackled using this method, plus scenes with fast camera motion i.e. very less overlapping between successive frames. Frames with insufficient trajectories are also bottle- neck for this method. Detailed comparison of stabilizing techniques is provided in table 3.

TABLE 3. COMPARISON OF EXISTING STABILIZING TECHNIQUES

Ref	Motion Model	Motion Estimation Technique	Complexity	Offline/ Online	Parallax Handling
[1]	Linear	SIFT Feature Tracking	Medium	Offline	No
[4]	2D	KLT Feature Tracking	Medium	Offline	Yes
[8,9]	2D Affine	HARIS Corner Detection and Tracking using KLT	Low	Online	Yes (Medium)
[10]	2D Affine	Steady Flow (Optical Flow)	Medium	Offline	Yes (Medium)
[11]	-	Phase Correlation	High	Offline	No
[12]	Perspective Projection	KLT Feature Tracking	Medium	Offline	No
[13]	2D Affine	Block Based Matching	High	Offline	No
[14]	2D Affine	SIFT Feature Tracking	High	Offline	No
[15]	Similarity	Circular Block Matching	Medium	Offline	No
[16]	2D Affine	SIFT and Particle Filters	High	Offline	Yes (Medium)
[17]	2D Affine	KLT Feature Tracking	High	Offline	Yes (High)
[18]	2.5 D	KLT Feature Tracking and Epipolar Lines	High	Offline	Yes (High)

IV. DISCUSSION

Though variety of mature video stabilization techniques have been developed in the past leading to development of some well-known commercial software's, still there exist some challenges that are yet to be undertaken. Most of the available image stabilizing techniques are limited by number of assumptions such as no moving object in center part of frame, tracking of road lines, preprocessing of feature trajectories (which is not feasible for real time processing), presence of camera at front of vehicles, high number of feature points throughout the video sequence etc. Available techniques perform well on variety of videos but in many challenging scenarios they simply fail. These scenarios include large depth variation, quick camera motion, large moving foreground, motion blur, rolling shutter effects etc. In real time stabilization algorithms future state of motion is unknown so most of existing techniques don't achieve notable performance in finding intentional motion, a robust intentional model is also needed which can handle all of possible intentional motions like panning, dolly motion, zoom. Most of the existing methods that achieve prominent performance have high computational cost and they are implemented as offline techniques. A robust and generic method is still required that can provide real time video stabilization along with parallax handling and should compensate maximum number of challenges mentioned above.

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